

THE LAYERS OF THE CONSECUTIVE-PRIME ARTIN CORRELATION: ONLY THE ABELIAN ONE CORRELATES

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ABSTRACT. The primitive-root (“Artin”) condition for a base a at a prime p factors into layers: for each prime $\ell \mid p-1$, the element a must not be an ℓ -th power residue. Earlier papers in this series measured a systematic correlation between the Artin statuses of consecutive primes and traced it, empirically, to the quadratic layer and to the joint distribution of $p-1$ ’s factorizations. This paper closes the remaining question: do the higher layers ($\ell \geq 3$) contribute correlation of their own? We show that they cannot contribute through gap classes — for $\ell \geq 3$ the ℓ -th power residue symbol of a fixed base is not determined by any congruence condition on p , because $\mathbb{Q}(\zeta_\ell, a^{1/\ell})$ is non-abelian over $\mathbb{Q}$, so no analogue of the quadratic exclusion laws exists — and we verify computationally that they do not contribute at all: over all consecutive prime pairs below 10^9 where the ℓ -th condition binds at both primes, the ℓ -th symbol statuses are statistically independent for every $\ell \in \{3, 5, 7, 11, 13\}$ and every base tested, while the *binding events* $\ell \mid p_n - 1$ and $\ell \mid p_{n+1} - 1$ are strongly anticorrelated (e.g. $\phi = -0.133$, $z = -150$, for $\ell = 3$ at 2×10^7) — the Lemke Oliver–Soundararajan bias modulo ℓ . The consecutive-prime Artin correlation is therefore carried entirely by the abelian ($\ell = 2$) layer and by the binding structure of $p-1$: a dichotomy aligned exactly with the abelian/non-abelian dichotomy of the Kummer extensions. As a corollary, the conditional model of the sixth paper in this series, which uses only quadratic and binding data, is not an approximation that happens to work: no further layer terms exist to add.

1. INTRODUCTION

Let $a \geq 2$ be a non-square integer. A prime $p \nmid a$ has a as a primitive root if and only if, for every prime $\ell \mid p-1$, the element a is not an ℓ -th power residue modulo p :

$$p \in A_a \iff a^{(p-1)/\ell} \not\equiv 1 \pmod{p} \quad \text{for every prime } \ell \mid p-1. \quad (1)$$

We refer to the condition at ℓ as the ℓ -th *layer*. The layer at ℓ *binds* at p when $\ell \mid p-1$; the layer at $\ell = 2$ binds at every odd prime, and its condition is exactly $\chi_a(p) = -1$ for the quadratic character $\chi_a = \left(\frac{\text{sqf}(a)}{\cdot}\right)$.

Earlier papers in this series measured the correlation

$$\delta(a; x) = P(p_{n+1} \in A_a \mid p_n \in A_a) - P(p_{n+1} \in A_a \mid p_n \notin A_a)$$

over consecutive primes up to $x = 10^9$ and found it nonzero for every base tested ($|z|$ up to 479), organised by the conductor of χ_a , with unconditional *exclusion laws* in the quadratic layer: gap classes modulo the conductor in which two consecutive primes can never both be Artin. A companion paper constructed a conditional Hardy–Littlewood model from the quadratic layer and the factorization structure of $p-1$ alone, and found it reproduces $\delta(a)$ to within 4% for twelve bases, with channel-level $R^2 > 0.999$.

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That success poses a question the model cannot answer for itself: *why is nothing missing?* The Artin condition (1) involves every layer. If the cubic layer, say, carried its own consecutive-prime correlation — an analogue of the quadratic exclusion laws, or any gap-class structure — the model would be incomplete and its agreement partly accidental. The present paper shows that the higher layers can carry no such structure, in two steps:

- **No exclusion laws above $\ell = 2$** (Section 2). The quadratic exclusion laws exist because $\chi_a(p)$ is periodic in p . For $\ell \geq 3$ the ℓ -th power residue status of a fixed integer a is *not* a congruence condition on p : the splitting field $\mathbb{Q}(\zeta_\ell, a^{1/\ell})$ is non-abelian over $\mathbb{Q}$, and a Chebotarev-style argument shows its splitting behaviour cannot be captured by residue classes. Gauss’s classical criterion — 2 is a cubic residue mod p iff $p = L^2 + 27M^2$ — is the concrete face of this obstruction.
- **Empirically, no correlation at all** (Section 3). Across all consecutive prime pairs below 10^9 at which the ℓ -th layer binds at both primes, the symbol statuses are statistically independent, for every $\ell \in \{3, 5, 7, 11, 13\}$ and every base $a \in \{2, 3, 5, 6, 7, 10, 11, 13, 15\}$. The *binding events* themselves, by contrast, are strongly anticorrelated — pure Lemke–Oliver–Soundararajan bias modulo ℓ [4].

The conclusion (Section 4) is a dichotomy that matches the Galois theory exactly: *the abelian layer correlates; the non-abelian layers are independent*. All consecutive-prime structure in the Artin property flows through (i) the quadratic character and (ii) which layers bind — both of which are congruence data — and none through the non-abelian symbols. In particular the companion model’s completeness is structural, not fortuitous.

2. NO EXCLUSION LAWS ABOVE THE QUADRATIC LAYER

The quadratic exclusion laws of the second paper rest on one fact: χ_a is a Dirichlet character modulo the conductor f , so $\chi_a(p)$ depends only on $p \bmod f$, and a gap class that forces $\chi_a(q) = -\chi_a(p)$ forces one of the two primes out of A_a . The following proposition shows this mechanism is exclusive to $\ell = 2$.

Proposition 1. *Let $\ell \geq 3$ be prime and let $a \geq 2$ be an integer that is not a perfect ℓ -th power. There is no modulus M and set $S \subseteq (\mathbb{Z}/M\mathbb{Z})^\times$ such that, for all sufficiently large primes $p \equiv 1 \pmod{\ell}$,*

$$a \text{ is an } \ell\text{-th power residue mod } p \iff p \bmod M \in S.$$

Proof. Let $K = \mathbb{Q}(\zeta_\ell, a^{1/\ell})$. For a prime $p \equiv 1 \pmod{\ell}$ not dividing $a\ell$, the condition “ a is an ℓ -th power residue mod p ” is equivalent to p splitting completely in K (standard Kummer theory: p splits completely in $\mathbb{Q}(\zeta_\ell)$ because $p \equiv 1 \pmod{\ell}$, and then splits completely in K iff $x^\ell - a$ has a root — equivalently all roots — mod p).

Suppose such M and S existed. Then the set of rational primes splitting completely in K would be, up to finitely many exceptions, a union of arithmetic progressions modulo $\text{lcm}(M, \ell)$. A number field whose set of completely-split primes is (up to density-zero exceptions) a union of arithmetic progressions mod N has the same completely-split set as some subfield of $\mathbb{Q}(\zeta_N)$; since the completely-split set determines the Galois closure (a standard consequence of the Chebotarev density theorem), the Galois closure of K would be contained in $\mathbb{Q}(\zeta_N)$ and hence *abelian* over $\mathbb{Q}$.

But K is its own Galois closure ($\zeta_\ell \in K$), with $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/\ell\mathbb{Z} \rtimes (\mathbb{Z}/\ell\mathbb{Z})^\times$ of order $\ell(\ell-1)$: it is non-abelian for every $\ell \geq 3$, because the subgroup fixing $\mathbb{Q}(\zeta_\ell)$ (translations $a^{1/\ell} \mapsto \zeta_\ell^j a^{1/\ell}$) is normal and acted on nontrivially by $(\mathbb{Z}/\ell\mathbb{Z})^\times$, which has order $\ell-1 \geq 2$. (Non-degeneracy of the action requires a not an ℓ -th power, which we assumed; the semidirect product is then genuinely non-abelian.) Contradiction. $\square$

TABLE 1. Binding correlations at $x = 10^9$: ϕ for the events $\ell \mid p_n - 1$ versus $\ell \mid p_{n+1} - 1$ over all consecutive pairs. All entries are exact counts over 50,847,527 pairs.

ℓ	ϕ	z
3	-0.10977	-782.7
5	-0.08909	-635.2
7	-0.07884	-562.2
11	-0.05954	-424.6
13	-0.05763	-411.0

Corollary 2. *For $\ell \geq 3$ there is no gap-class exclusion law at the ℓ -th layer: no modulus M and residue class g_0 such that consecutive primes $p, q = p + g$ with $g \equiv g_0 \pmod{M}$, both $\equiv 1 \pmod{\ell}$, are forbidden from both having a as an ℓ -th power non-residue (or any other fixed pair of symbol statuses).*

Proof. A gap-class law forcing symbol statuses would, applied to the pairs $(p, p + g)$ ranging over all admissible p , determine the symbol status of $p + g$ from congruence data of p and g alone — in particular the symbol status at a prime q would be a congruence condition on q (intersect over the available p ; the LOS equidistribution of p over admissible classes makes every class available). This contradicts Proposition 1. $\square$

Remark 3. The classical criteria make the obstruction tangible. 2 is a cubic residue modulo $p \equiv 1 \pmod{3}$ iff $p = L^2 + 27M^2$ (Gauss); 2 is a quartic residue modulo $p \equiv 1 \pmod{8}$ iff $p = L^2 + 64M^2$. Representation by a quadratic form of discriminant -108 or -256 is class-field data over $\mathbb{Q}(\zeta_3)$ resp. $\mathbb{Q}(i)$ — abelian over the quadratic field, but not over $\mathbb{Q}$ — and is well known not to reduce to congruences in p .

Remark 4. Proposition 1 does not by itself force statistical independence of the symbols at consecutive primes; it only removes the congruence mechanism that produces exclusion laws and, more generally, any correlation *mediated by joint residue classes* of (p_n, p_{n+1}, g) — which is precisely the mechanism (Hardy–Littlewood singular series, LOS bias) known to govern consecutive-prime dependence. Whether some other mechanism correlates the non-abelian symbols across consecutive primes is an empirical question, which we now settle at scale.

3. MEASUREMENT: THE HIGHER SYMBOLS ARE INDEPENDENT

3.1. Setup. For each prime $\ell \in \{3, 5, 7, 11, 13\}$ and base $a \in \{2, 3, 5, 6, 7, 10, 11, 13, 15\}$, define at each prime p with $\ell \mid p - 1$ the symbol status $s_\ell(a, p) = 1$ if $a^{(p-1)/\ell} \equiv 1 \pmod{p}$, else 0. We accumulate, over consecutive prime pairs (p_n, p_{n+1}) below 10^9 : (i) the 2×2 table of binding events $(\ell \mid p_n - 1, \ell \mid p_{n+1} - 1)$; (ii) for pairs binding at both primes, the 2×2 table of $(s_\ell(a, p_n), s_\ell(a, p_{n+1}))$; (iii) for the $\ell = 2$ layer, the table of quadratic-character signs; and (iv) the Artin-status table, as a consistency anchor to the earlier papers. We report the ϕ coefficient (Pearson correlation of the two indicators) and $z = \phi\sqrt{n}$.

The computation factors each $p - 1$ once and evaluates all layers and bases in a single pass (the scheme of the second paper); a full run to 10^9 takes about an hour on one core. The pilot stage was implemented independently in `sympy` and agrees with the C implementation to all reported digits at 2×10^7 .

TABLE 2. Symbol correlations at $x = 10^9$, conditional on binding at both primes: ϕ (with z in parentheses) for $s_\ell(a, p_n)$ versus $s_\ell(a, p_{n+1})$. Maximum $|z|$ over all 45 combinations: 2.4, consistent with 45 null tests.

a	$\ell = 3$	$\ell = 5$	$\ell = 7$	$\ell = 11$	$\ell = 13$
2	−0.00024 (−0.8)	+0.00080 (+1.2)	−0.00034 (−0.3)	−0.00028 (−0.1)	−0.00077 (−0.3)
3	+0.00015 (+0.5)	+0.00047 (+0.7)	+0.00150 (+1.4)	−0.00137 (−0.7)	−0.00020 (−0.1)
5	−0.00008 (−0.3)	−0.00140 (−2.1)	+0.00035 (+0.3)	+0.00155 (+0.8)	+0.00192 (+0.7)
6	−0.00002 (−0.1)	+0.00031 (+0.5)	+0.00000 (+0.0)	+0.00413 (+2.0)	+0.00141 (+0.5)
7	−0.00026 (−0.9)	+0.00063 (+1.0)	+0.00045 (+0.4)	+0.00179 (+0.9)	+0.00061 (+0.2)
10	+0.00023 (+0.8)	+0.00080 (+1.2)	+0.00237 (+2.2)	+0.00138 (+0.7)	−0.00051 (−0.2)
11	−0.00031 (−1.0)	+0.00017 (+0.3)	−0.00174 (−1.6)	−0.00418 (−2.0)	−0.00302 (−1.1)
13	−0.00012 (−0.4)	+0.00011 (+0.2)	−0.00126 (−1.2)	−0.00084 (−0.4)	−0.00023 (−0.1)
15	−0.00027 (−0.9)	−0.00002 (−0.0)	−0.00124 (−1.1)	−0.00207 (−1.0)	+0.00668 (+2.4)

3.2. Results. Two facts, in deliberate contrast:

Observation 5 (Binding layer: strong anticorrelation). The binding events are anticorrelated for every ℓ , at overwhelming significance. This is the Lemke Oliver–Soundararajan bias modulo ℓ in its purest form: a prime $\equiv 1 \pmod{\ell}$ repels a successor in its own residue class, and $1 \pmod{\ell}$ is a single class among $\ell - 1$.

Observation 6 (Symbol layer: independence). Conditional on binding at both primes, the symbol statuses are independent: across all 45 (base, ℓ) combinations, no $|z|$ exceeds the range expected from 45 null tests. The marginal symbol frequencies are $1/\ell$ to high accuracy, as they must be (Kummer–Chebotarev equidistribution), and the joint frequencies factor.

3.3. Consistency anchor. The same run reproduces the Artin-status correlations $\delta(a)$ of the earlier papers to the last digit for all nine bases (e.g. $\delta(2) = -0.050199$, $\delta(15) = +0.012196$, matching to six decimals), confirming that the layer data and the Artin data come from the identical prime stream.

4. DISCUSSION

The dichotomy. Assemble the three findings: (i) the quadratic symbol is periodic in p , inherits the consecutive-prime residue bias, and produces exclusion laws and the conductor-organised correlation of the earlier papers; (ii) the binding events at every ℓ are congruence data ($p \pmod{\ell}$) and are strongly anticorrelated; (iii) the non-abelian symbols at every $\ell \geq 3$ carry no consecutive-prime correlation at all. The split is exactly the abelian/non-abelian split of the splitting fields: $\mathbb{Q}(\sqrt{\text{sqf } a})$ and $\mathbb{Q}(\zeta_\ell)$ are abelian over $\mathbb{Q}$, so their splitting data are congruence conditions, and consecutive primes are biased in congruence data; $\mathbb{Q}(\zeta_\ell, a^{1/\ell})$ for $\ell \geq 3$ is non-abelian, its splitting data are invisible to residue classes, and consecutive primes appear to carry no bias in them whatsoever.

We find the cleanliness of (iii) worth emphasising. Nothing in Proposition 1 *forbids* a correlation of non-abelian symbols — it only rules out the congruence mechanism. The Hardy–Littlewood/LOS framework, as far as it is understood, biases exactly the joint congruence data of consecutive primes; our measurement is consistent with the sharpest reading of that framework: *consecutive primes are dependent only through congruence data*. Symbol independence at 45 combinations, with the binding channels simultaneously showing $|z| > 100$ in the same

data stream, is a strong internal control: the pipeline demonstrably detects correlation where correlation exists.

Consequences for the series. First, the conditional model of the sixth paper is structurally complete: its inputs (quadratic layer, binding structure, gap classes) exhaust the channels through which consecutive-prime dependence can enter the Artin property, so its $R^2 > 0.999$ channel agreement is not a fortunate truncation. Second, the “cubic or quartic sharing” speculation recorded in the third paper as a possible source of a small residual floor is now implausible: the cubic symbols themselves are uncorrelated, so any residual in that paper’s cross-base analysis must come from the binding structure or from finite-size effects, not from a cubic analogue of quadratic entanglement. Third, the exclusion-law programme of the second paper is complete as stated: there are no higher-layer exclusion laws to find.

Limitations. The independence claim is statistical, bounded by the resolution of the census ($\approx 6 \times 10^{-4}$ in ϕ at $z = 4$ for $\ell = 3$; weaker for larger ℓ , where binding pairs are scarcer). A correlation of order 10^{-4} at $\ell = 3$ would evade us. We see no mechanism that would produce one, but the bound is what the data give. Second, Proposition 1 is stated for prime ℓ and non- ℓ -th-power a ; composite indices add nothing new (the layers of (1) are prime), but bases that are perfect ℓ -th powers have degenerate layers and are excluded throughout, as in the earlier papers.

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DECLARATION OF INTEREST

The author reports there are no competing interests to declare.

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DATA AVAILABILITY

All code and results are publicly available at <https://github.com/jpbald93/consecutive-artin>, including the one-pass layer census (C), the independent `sympy` pilot, the exact contingency tables for every (base, ℓ) combination, and the scripts reproducing every table.

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